EXPLAINABLE AI: TECHNIQUES AND APPLICATIONS WITH ML IN EDUCATION TECHNOLOGY

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Abstract

Educational technology and explainable AI research papers have gained significant attention in recent years. Researchers have utilized artificial intelligence (AI) in education to develop intelligent tutoring and learning systems. A comprehensive review of empirical studies on AI in education highlights the current state of AIEd research, its technologies, applications, and benefits for education. Content analysis of studies on AI in education reveals research trends such as classification, matching, recommendation, deep learning, feedback, reasoning, adaptive learning, and integration of AI in education. Exponential technology involves AI Literacy, LMS Moodle, Results Analytics, we need such terminologies for this research. Future developments in educational technology and AI in education present challenges related to intercultural dimensions, privacy, collaboration at scale, and the role of AIED In educational technology. *Keywords*: Artificial Intelligence, Explainable AI, Exponential Techniques, Education Technology, Machine Leaning, Applications, Black box, Results Analytics, LMS Moodle.

I. Introduction

An emerging field called Explainable AI (XAI) seeks to improve the transparency and comprehension of machine learning models, particularly in the context of educational technology.Machine learning is a ke y component of personalised learning, adaptive assessments, and intelligent tutoring systems in the field of education.The blackbox aspect of these models, however, can undermine confidence and reduce their teaching potential.To close this gap, experiential methods and XAI applications in educational technolog y are crucial.

These methods give instructors, pupils, and educational institutions a better understanding of how AI alg orithms function and make judgement calls.

Understanding the rationale behind recommendations allows educators to modify their lesson plans and i nstructional techniques to better meet the requirements of specific stuadents. Students can better grasp their own learning progress thanks to XAI's explanations for AI-generated recommendations and assessments. Additionally, by ensuring fairness and equity in their algorithms, educational institutions can lessen bias and discrimination. This article examines several XAI applications in education technology that are really being used, emphasising how these techniques enable stakeholders to fully utilise machine learning while promoting a more open and accountable learning environment. In the end, XAI in educational technology has the promise of enhancing student engagement, personalised learning, Results Analytics, LMS Moodle, and learning outcomes. Fig 1 shows Synergy between various stakeholders in XAI.

II. Literature Review

Artificial Intelligence

According to Verma (2018), the interdisciplinary field of artificial intelligence (AI) aims to develop smart computers that are capable of doing activities that typically require human intelligence. Among these tasks are natural language understanding, pattern identification, decision-making, and experience-based learning. Some of the subfields of AI that it embraces include machine learning, natural language processing, computer vision, robotics, and expert systems. The ultimate goal of AI is to create machines that can duplicate or outperform human intelligence in specific disciplines.

A literature review provides a summary of the state of the art in artificial intelligence knowledge and research. It is necessary to analyze and synthesize pertinent academic books, journals, conference proceedings, and other trustworthy sources.

Explainable AI

The development of artificial intelligence systems that offer clear and intelligible justifications for their judgements and forecasts is known as explainable AI (XAI). By making it possible for people to understand how and why AI systems make particular decisions, it strives to overcome the "black box" aspect of complicated AI models. Model visualisation, feature importance analysis, and explanations in natural language are all examples of XAI approaches. In industries like healthcare, banking, and autonomous cars, where decisions have substantial real-world repercussions, achieving XAI is essential for improving trust, accountability, and regulatory compliance in AI applications. It makes ensuring AI adheres to human values and expectations and behaves accordingly.

Education Technology

The use of digital tools and resources in the learning process is known as educational technology, or EdTech. By offering a variety of chances for individualised, interactive, and accessible learning experiences, it has revolutionised education. It consists of a variety of parts, such as adaptive learning systems, instructional software, and online learning platforms. Remote learning is made easier and more accessible to a worldwide audience thanks to edtech. To engage students and enhance learning outcomes, it also uses technology like artificial intelligence, augmented reality, and gamification. Additionally, EdTech provides teachers with resources for data-driven instruction, empowering them to adapt their lesson plans and curricula to each student's requirements. EdTech has enormous potential to transform education's future and make it more inclusive, engaging, and efficient as technology develops.

Machine Learning

According to Hajdegani Razieh, the models and techniques that enable computers to learn from data are the main topics of machine learning (ML) literature. It covers key concepts including supervised learning, where models are trained using labelled samples, and unsupervised learning, which focuses on detecting patterns and structures in unlabeled data. Reinforcement learning is a significant subject as it focuses on learning through interaction with the environment. Reinforcement learning is a subsection of machine learning.

Expert Systems

J. Clerk Maxwell (1992) said that expert systems are Artificial Intelligence (AI) tools that use logic and subject-matter knowledge to solve complex problems in certain disciplines. It consists of an inference engine that uses logical reasoning to draw inferences or provide decisions based on the information presented and a knowledge base that holds pertinent information and guidelines [A]. Expert systems, which imitate the knowledge of human professionals and offer insightful analysis, ideas, and problem-solving abilities, can assist in fields like medicine, engineering, finance, and troubleshooting. These systems are able to give reliable knowledge across a variety of disciplines, boost effectiveness, and speed up decision-making.

Explainable Technology

Explainable Technology refers to its effects in the applied research to this paper. Specifically, the benefits to the people involved in the research model, how to deliver its applicability, what are the various models of ML used in education, the human centered design of Artificial Intelligence and potential differences/ non negotiable terms/ pitfalls of the model. Also, AI Literacy is equally important in terms of using technology at its supreme level and making technology ubiquitous referred in figure 2.

III. RESEARCH METHODOLOGY

In research methodology, specifically there are five questions, which seek answers, through this paper, as follows:

1. How can XAI techniques be applied to improve the transparency and interpretability of ML models used in Educational settings?

2. List some common challenges and limitations associated with implementing XAI in Educational ML Systems.

3. How does XAI contribute to personalized learning in education and what benefits does it offer to the students and educators?

4. What role can XAI play in addressing bias and fairness issues in educational ML Algorithms such as in grading or admissions process?

5. How can educators and students be trained or educated about XAI to better understand and trust the AI driven tools they use?

Q1. How can XAI techniques be applied to improve the transparency and interpretability of ML models used in Educational settings?

ML models employed in educational contexts can be made more transparent and interpretable by using explainable AI (XAI) approaches. In order to increase the trustworthiness and comprehension of ML models, XAI methods seek to offer insights into the decision-making process. Rick Willming in 2023, stated that with the aid of these tools, educators can better comprehend the underlying assumptions that lead to the models' projections. Healthcare professionals can rely on and use XAI to efficiently employ these models for precise and trustworthy medical diagnosis. The effectiveness of XAI techniques has also been examined in the clinical setting, where risk prediction models were examined using the justifications produced by these techniques.

The findings draw attention to the drawbacks of current XAI methodologies in the clinical setting while also highlighting their potential use in assisting with clinical decision-making. Additionally, XAI can be utilised to provide transparency and dependability in decision-making by outlining the justification for the decisions made by ML algorithms. Rene' F. Kizilcec 2023, has also suggested to use XAI approaches, such as Genetic Programming Explainer (GPX), to explain the choices made by AI systems. Interpretability is improved by GPX, which creates understandable analytical expressions that capture the local behaviour of complex models.

Q2. List some common challenges and limitations associated with implementing XAI in Educational ML Systems.

XAI implementation in ML systems for education is hampered by a number of issues. Lack of transparency in AI systems is a prevalent issue that makes it challenging for users to understand the system's output. Particularly in sensitive fields like medicine, this lack of transparency can cause problems with under- or over-reliance on the system's suggestions. According to Aida Brankovic 2023, making ethical and equitable decisions is a difficulty as well because AI that has been educated on previous data might repeat past behaviours and prejudices. Along with a severe absence of user studies examining the demands of physicians, there is also a distinct lack of XAI application in the context of clinical decision support systems (CDSS). These difficulties emphasise the significance of including XAI in educational ML systems to ensure transparency, fairness, and trustworthiness.

Q3. . How does XAI contribute to personalized learning in education and what benefits does it offer to the students and educators?

By offering specialized learning materials and recommendations based on personal preferences, AI supports personalized learning in education and meets the varied needs of students [1]. Both educators and students can gain from it. AI systems can enhance the learning experience for students by scaling up personalized learner-instructor interaction, delivering just-in-time, personalized support, and enhancing the sense of community [2]. Better learning results and higher levels of satisfaction may result from this. AI technologies for educators can improve performance, streamline administrative work, and provide them more time and freedom to concentrate on talents that are uniquely human, including learning and adapting. Jeff Druce 2021, noted the advantages of one-to-one learning, which has been shown to

considerably boost student performance, may also be unlocked by AI-driven personalized learning apps. AI technologies can improve educational fairness, academic success, and learning outcomes by comprehending the needs and viewpoints of educators. Figure 3 clearly demonstrates the relationships between various entitites.

Q4. What role can XAI play in addressing bias and fairness issues in educational ML Algorithms such as in grading or admissions process?

In educational ML algorithms like grading or admissions procedures, prejudice and fairness issues can be greatly reduced with the use of explainable artificial intelligence (XAI). XAI offers transparency and aids in comprehending the justifications for the choices made by ML algorithms. As per Utsab Khakurel 2022, fair results are ensured by allowing for the detection and mitigation of biases in the data. In order to identify potential biases and their influence on the algorithm's decisions, XAI can assist in examining the impact of various features on the prediction process. Making educational ML algorithms more dependable and transparent with XAI approaches will ensure that admissions or grading procedures are impartial and fair.

In order to reduce any algorithmic unfairness, XAI can also assist in evaluating the fairness measures of ML algorithms and putting de-biasing strategies into practice.

Q5 How can educators and students be trained or educated about XAI to better understand and trust the AI driven tools they use?

Amiransalan Rajabi 2021, concludes that through proactive measures taken by educators, teachers and students can be trained or educated about XAI to better comprehend and trust AI-driven systems. By emphasizing its useful applications and encouraging AI literacy, educators might assist students in reorienting their thinking about AI. Tommy van der Vorst 2019, stated that students will learn how to use AI tools to become resourceful mathematicians, and the value of maintaining the information in reliable sources will also be emphasized . As per Leonardo Augusto 2020, educators can also serve as knowledge curators by sharing their own real-world experience with pupils and assembling instructional resources from reliable sources . F. Kearns Charles 2023 demonstrated that to teach students how to utilize these tools effectively, it is crucial to connect with and foster AI literacy. Teachers can contribute to the continuing discussion on XAI by fostering interpretability and transparency in AI systems.

CONCLUSION

Thus, the research paper can be considered to highlight the importance and benefits of explainable AI in the field of education technology, suggesting the growing recognition of the need for transparency and interpretability in AI models used for educational purposes. The mention of interdisciplinary nature of AI and its subfields indicates that AI is a broad field with various applications beyond education. This paper serves to be an acknowledgement of the existence of challenges and limitations in implementing XAI in educational ML systems, implying that while XAI is promising, there stand practical hurdles that need to be overcome to effectively integrate it into educational technology. The use of expert systems in problem-solving, implies that AI techniques like expert systems are being explored in the context of education technology and may be found to be constant feature in the field of education in the coming years, provided that we find ways to counter the obstacles in the field of XAI's integration.

Both educators and students can gain from AI because, by offering specialized learning materials and recommendations based on personal preferences, AI supports personalized learning in education and meets the varied needs of students. AI systems can enhance the learning experience for students by scaling up personalized learner-instructor interaction, delivering just-in-time, personalized support, and enhancing the sense of community. This furthers not only into better learning results but also higher levels of satisfaction. AI technologies for educators can improve performance, streamline administrative work, and provide them more time and freedom to concentrate on talents that are uniquely human, including

learning and adapting. The advantages of one-to-one learning, which has been shown to considerably boost student performance, may also be unlocked by AI-driven personalized learning apps. In educational ML algorithms like grading or admissions procedures, prejudice and fairness issues can be greatly reduced with the use of explainable artificial intelligence (XAI). Fair results are ensured by allowing for the detection and mitigation of biases in the data.

FUTURE SCOPE AND LIMITATIONS

In order to identify potential biases and their influence on the algorithm's decisions, XAI can assist in examining the impact of various features on the prediction process. Making educational ML algorithms more dependable and transparent with XAI approaches will ensure that admissions or grading procedures are impartial and fair. By emphasizing its useful applications and encouraging AI literacy, educators might assist students in reorienting their thinking about AI. Educating teachers and students about XAI to better comprehend and trust AI-driven systems may lead to better educational processes and an overall finetuned educational society.

Future Scope lies in further catering to individual needs of students, their educational goals, objectives, adjusting educational materials to the needs of particular students, assisting teachers in creating more efficient curriculum, and spotting warning indications of student difficulty. In order to increase trust and openness in AI-driven educational systems, XAI is crucial. The demand for XAI techniques that not only offer precise forecasts and suggestions but also explain the reasoning behind them will increase as the use of AI-powered tools in education increases. This will enable educators and students to make more educated decisions. Furthermore, ML and XAI can help automate administrative work so that teachers can concentrate more on teaching and mentoring. The use of ML and XAI in education will depend on developing flexible, open, and egalitarian learning environments that fully utilise AI while preserving moral principles and encouraging diversity.

Figures and Tables:



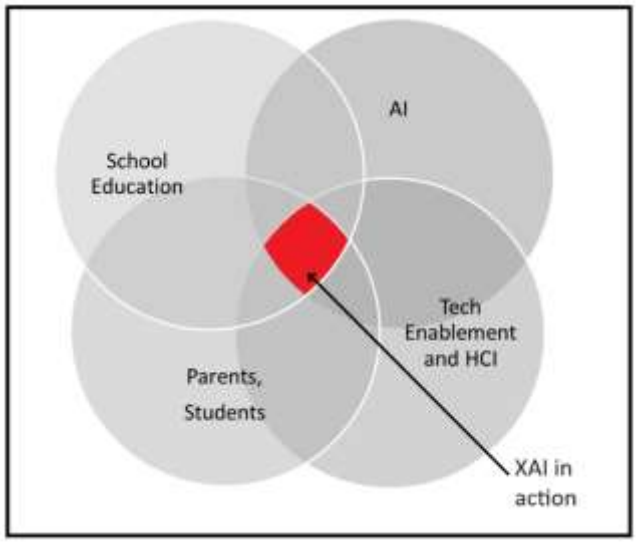


Fig 1: Synergy between various stakeholders in XAI Fig 2: XAI sweetspot, in working mode

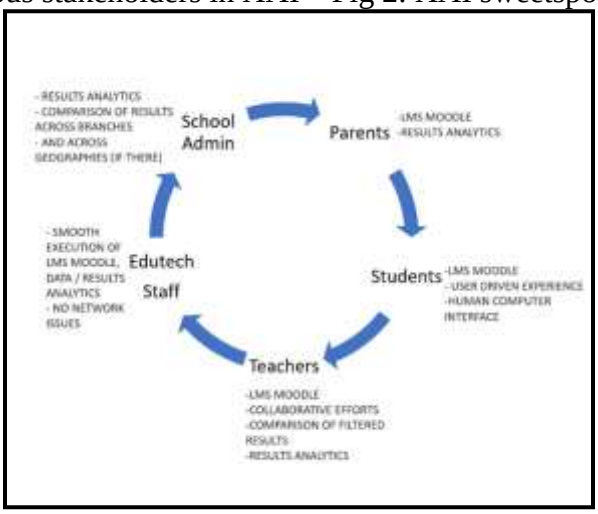


Figure 3: Various aspects of Interim Stakeholders/ Key decisionmakers of the educational ecosystem.

References

- Aida Brankovic, David Cook, Sankalp Khanna , 21 Jun 2023, Evaluation of Popular XAI Applied to Clinical Prediction Models: Can They be Trusted?, arXiv.org, Vol. abs/2306.11985(10)
- Amirarsalan Rajabi1, Ozlem Ozmen Garibay, 24 Jul 2021, Towards Fairness in AI: Addressing Bias in Data Using GANs, University of Central Florida, -pp 509-518(18)
- F Kearns Charles, 23 May 2023, AI-Powered Personalized Mobile Education for New Zealand Students,International journal of technology, innovation and management (International journal of technology, innovation and management)-Vol. 3, Iss: 1, pp 43-49(16)
- Hajdegani Razieh, et al, A Review on Expert System in Medicine(2)
- I. S. Jacobs and C. P. Bean, , 1963, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: -Academic, pp. 271–350.(4)
- J. Clerk Maxwell, 1892, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon , pp.68–73.(3)

- Jeff Druce, James Niehaus, Vanessa Moody, David Jensen, Michael L. Littman Brittle, 10 Jun 2021, AI, Causal Confusion, and Bad Mental Models: Challenges and Successes in the XAI Program. arXiv: Artificial Intelligence-(15)
- K. Elissa, “Title of paper if known,” unpublished.(5)
- Leonardo Augusto Ferreira1, Frederico Gadelha Guimarães, 19 Jul 2020,Applying Genetic Programming to Improve Interpretability in Machine Learning Models, , Rodrigo Silva2•Institutions (2), , pp 1-8(12)
- M. Young, The Technical Writer’s Handbook. Mill Valley, 1989, CA: University Science.(8)
- Faraón Llorens Largo, Leandro Tortosa, Jose F. Vicent , 01 Oct 2023,Explainability techniques applied to road traffic forecasting using Graph Neural Network models, Information Sciences, Vol. 645, pp 119320-119320(9)
- René F. Kizilcec, 09 Jun 2023 Bias, Fairness, and Accountability with AI and ML Algorithms To Advance AI Use in Education, Focus on Understanding Educators (19)
- Ram Machlev, Leena Heistrene, Michael W. Perl, Kedem Levy, J.A. Belikov, S. Mannor, Yoash Levron , May 2022, Explainable Artificial Intelligence (XAI) techniques for energy and power systems: Review, challenges and opportunities,Energy and AI-Vol. 9, pp 100169-100169(14)
- R. Nicole, “Title of paper with only first word capitalized,” J. Name Stand. Abbrev., in press.(6)
- Rick Wilming, Stefan Haufe un , 2023 , XAI-TRIS: Non-linear benchmarks to quantify ML explanation performance, arXiv.org, Vol. abs/2306.12816(13)
- Tommy van der Vorst, Nick Jelicic, 01 Jan 2019, Artificial Intelligence in Education: Can AI bring the full potential of personalized learning to education?(17)
- Utsab Khakurel, Danda B. Rawat, 06 Jun 2022, Evaluating explainable artificial intelligence (XAI): algorithmic explanations for transparency and trustworthiness of ML algorithms and AI systems, Vol. 12113, pp 121130A-121130A(11)
- Verma Mudit, AI and its application with special emphasis in education, Mudit Verma. (2018). International Journal of Advanced Educational Research.(1)